



FINANCIAL MATHS

PERPETUITY | SINKING FUND | EMI — BY THUNDERSTUDY



1. PERPETUITY (Money Forever)

Money paid infinitely. Keywords: "Forever", "Infinite", "Permanent".

$$\text{Present Value} = \text{Income} / \text{Rate}$$

MCQ Shortcut: Convert % to decimal (10% = 0.1) and just **DIVIDE**. Simple!



2. SINKING FUND (Future Goal)

Saving small deposits to reach a big target. Keywords: "Create Fund", "Accumulate".

$$\text{Deposit} = (\text{Future Sum} \times r) / [(1+r)^n - 1]$$

Calculator Hack: Find Numerator first. Then approximate the power (1.1^5 is approx 1.6). Pick the nearest option!



3. EMI (Loan Repayment)

Monthly loan installments. **Warning:** Use Monthly Rate!

$$\text{EMI} = [P \cdot r \cdot (1+r)^n] / [(1+r)^n - 1]$$

Shortcut: If rate = 12% yearly, use 1% (0.01) monthly. Approximate powers to eliminate extreme options.



ULTRA-FAST IDENTIFICATION HACK

Keyword Found	Formula / Chapter
"Paid Forever"	✓ Perpetuity
"Reach Future Target"	✓ Sinking Fund
"Loan Repayment"	✓ EMI

70% of MCQs are solved just by picking the right formula!

⚡ 10-Second MCQ Strategy

1. Read **LAST line** first to find what is asked.
2. Identify keyword (Forever / Future / Loan).
3. Convert rate strictly to decimal/monthly.
4. Approximate powers for rough division.
5. Choose the nearest option.

THUNDERSTUDY: MONEY MATHS MADE EASY